

Smart Contract Audit Report

MoonsaleTokenLock

BNB Smart Chain · Solidity ^0.8.20

REPORT ID	IGH-MSL-2026-009
CLIENT	Moonsale App
SCOPE	MoonsaleTokenLock.sol (third-round, L-02 + I-03)
VERSION	v3 (post-remediation of IGH-MSL-2026-008)
METHODOLOGY	Targeted diff review, regression scan
ISSUE DATE	April 23, 2026

Executive Summary

This is a targeted follow-up to IGH-MSL-2026-008. Two changes were applied to MoonsaleTokenLock.sol in this round: nonReentrant was added to sweepStuckETH (closing L-02 from the previous report), and a new extendLock function was introduced (closing I-03). Both changes verify as correctly implemented. No regressions were introduced. The score rises from **88 / 100** to **92 / 100**.

The extendLock implementation is notable because it proactively adopts the 3,650-day (10-year) upper bound on newUnlockTime that was recommended for the analogous function in MoonsaleFactory under IGH-MSL-2026-006 (factory L-03). The client applied the pattern here without being prompted, preventing the self-griefing footgun on day one.

Overall Security Score

92 / 100

was 88 / 100

Verification

L-02 Fix — sweepStuckETH nonReentrant

Function	Before (v2)	After (v3)
sweepStuckETH	external onlyOwner	external onlyOwner nonReentrant

Function body is byte-identical to v2. The modifier is correctly placed after onlyOwner. Coverage of nonReentrant across state-mutating external functions is now complete and uniform.

I-03 Fix — extendLock function added

The new function implementation:

```
function extendLock(uint256 lockId, uint256 newUnlockTime) external nonReentrant {
    Lock storage l = locks[lockId];
    require(l.owner == msg.sender, "Not lock owner");
    require(!l.withdrawn, "Already withdrawn");
    require(newUnlockTime > l.unlockTime, "Must be later than current unlock");
    require(newUnlockTime <= block.timestamp + 3650 days, "Exceeds max lock duration");
    l.unlockTime = newUnlockTime;
    emit LockExtended(lockId, newUnlockTime);
}
```

Edge cases traced:

- **Non-owner caller.** First require blocks. Correct.
- **Already withdrawn lock.** Second require blocks. The lock record still exists but l.amount was transferred out on withdraw, so extending a withdrawn lock would be meaningless. Correctly prevented.
- **Shortening attempt.** newUnlockTime > l.unlockTime is strict inequality, so the owner cannot shorten or even match the current unlock time. This closes a potential rug vector where a lock owner could appear to commit and then reduce the commitment.
- **Unbounded far-future extension.** The 3,650-day upper bound (approximately 10 years from the current block) prevents setting newUnlockTime to type(uint256).max or any similarly extreme value. This is the exact bound recommended for the factory's extendLock under IGH-MSL-2026-006.
- **Expired-but-not-withdrawn lock.** If the original unlock time has passed and the owner has not yet withdrawn, the lock can still be extended. This is correct behaviour: the lock record is still live and the owner

retains the right to re-commit.

- **nonReentrant.** The function makes no external calls, so the guard is not strictly necessary. Including it aligns with the contract's uniform convention and provides insurance against future edits that may add external calls.
- **Event.** `LockExtended(uint256 indexed lockId, uint256 newUnlockTime)` is declared and emitted. Indexers can track extensions cleanly.

Regression Scan

- Storage layout: unchanged.
- Constructor: unchanged.
- Existing functions (`createLock`, `withdraw`, view functions, admin setters): unchanged.
- H-01, M-01, M-02, L-01, L-02 (v1), L-04, L-05 fixes from IGH-MSL-2026-008: still in place, unaltered.
- New function is completely additive. No existing code paths touched.

Round-by-Round Status

ID	Finding (short)	Status
L-02	sweepStuckETH lacks nonReentrant	RESOLVED
I-03	No function to extend lock unlock time	RESOLVED
L-01	Reverting feeRecipient DoSes createLock	OPEN
I-01	Missing events for setFeeRecipient and transferOwnership	OPEN
I-02	LockCreated event omits description	OPEN
I-04	No recovery path for stranded tokens	OPEN

Lower-severity items from IGH-MSL-2026-008 that remain open: one Low (L-01, reverting recipient DoS) and three Informational (I-01, I-02, I-04). Descriptions and recommendations in the previous report still apply.

Reentrancy Coverage Matrix (post-v3)

Function	Guard	External call?
createLock	nonReentrant	Yes (fee fwd, excess refund, transferFrom)
withdraw	nonReentrant	Yes (transfer)
extendLock	nonReentrant (v3)	No
sweepStuckETH	nonReentrant (v3)	Yes (value transfer)
setFee, setFeeRecipient, setCreationPaused, transferOwnership	onlyOwner	No

Every external function that performs an external call carries nonReentrant. Coverage is complete and uniform.

Summary

Two clean, additive changes this round. L-02 closed by a single-keyword modifier addition to sweepStuckETH. I-03 closed by a well-designed extendLock function that adopts the factory's pattern including the 10-year upper bound. No regressions. The remaining open items are one Low (reverting recipient DoS) and three Informational, all carried forward from IGH-MSL-2026-008 with the client's stated plan to address them in final polish.

Deployment posture: production-ready. No blockers.

Security Score	92 / 100	0 High · 0 Medium · 1 Low · 3 Info
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Disclaimer

This report reflects a point-in-time third-round follow-up review of the Solidity source code provided by the client on the issue date shown on the cover. It covers only the changes made since IGH-MSL-2026-008; findings carried forward from that report retain their original descriptions and recommendations. A successful audit does not guarantee the absence of vulnerabilities, and changes to the code after review invalidate the conclusions. This report is an advisory document and does not constitute financial, investment, legal, or tax advice. ICOGemHunters accepts no liability for losses arising from the non-implementation of recommendations contained herein, from implementation errors introduced when applying recommendations, or from interactions with contracts outside the reviewed scope.

MoonsaleTokenLock has now completed three audit rounds. When L-01 or any of the remaining Informational items are addressed, a short targeted follow-up can be requested.